One-Step & Two-Step Equations

Practice Worksheet

Name: _____ Date: _____ Period: _____

Goal: Get x alone on one side by doing the **same operation to both sides**.

Operation	Undo it with...	Example
$x + a = b$	subtract a from both sides	$x + 3 = 7 \Rightarrow x = 4$
$x - a = b$	add a to both sides	$x - 5 = 2 \Rightarrow x = 7$
$ax = b$	divide both sides by a	$3x = 12 \Rightarrow x = 4$
$\frac{x}{a} = b$	multiply both sides by a	$\frac{x}{4} = 3 \Rightarrow x = 12$

Worked Examples

One-Step Example: Solve $x + 6 = 10$

Subtract 6 from both sides:

$$x + 6 - 6 = 10 - 6$$

$$\boxed{x = 4}$$

Check: $4 + 6 = 10$ ✓

Two-Step Example: Solve $3x - 5 = 7$

Step ① Add 5 to both sides:

$$3x - 5 + 5 = 7 + 5 \Rightarrow 3x = 12$$

Step ② Divide both sides by 3:

$$\frac{3x}{3} = \frac{12}{3} \Rightarrow \boxed{x = 4}$$

Check: $3(4) - 5 = 12 - 5 = 7$ ✓

Part A One-Step Equations (10 problems)

One operation separates x from the answer. Undo it and solve.

1. $x + 5 = 9$

Operation: _____

$x =$ _____

Check: _____

2. $x - 3 = 7$

Operation: _____

$x =$ _____

Check: _____

3. $2x = 14$

Operation: _____

$x =$ _____

Check: _____

4. $\frac{x}{4} = 3$

Operation: _____

$x =$ _____

Check: _____

5. $x + 8 = 2$

Operation: _____

$x =$ _____

Check: _____

6. $x - 6 = -1$

Operation: _____

$x =$ _____

Check: _____

7. $3x = -12$

Operation: _____

$x =$ _____

Check: _____

8. $\frac{x}{5} = -2$

Operation: _____

$x =$ _____

Check: _____

9. $-x = 9$

Hint: divide by -1

$x =$ _____

Check: _____

10. $x + 4 = -3$

Operation: _____

$x =$ _____

Check: _____

Part B Two-Step Equations (10 problems)

Step ①: Add or subtract to move the constant. *Step ②: Multiply or divide to isolate x.*

1. $2x + 3 = 11$
Step ①: $2x =$ _____
Step ②: $x =$ _____
Check: _____

2. $3x - 5 = 10$
Step ①: $3x =$ _____
Step ②: $x =$ _____
Check: _____

3. $4x + 1 = 17$
Step ①: $4x =$ _____
Step ②: $x =$ _____
Check: _____

4. $2x - 7 = 1$
Step ①: $2x =$ _____
Step ②: $x =$ _____
Check: _____

5. $5x + 2 = -8$
Step ①: $5x =$ _____
Step ②: $x =$ _____
Check: _____

6. $3x - 4 = -13$
Step ①: $3x =$ _____
Step ②: $x =$ _____
Check: _____

7. $\frac{x}{2} + 3 = 7$
Step ①: $\frac{x}{2} =$ _____
Step ②: $x =$ _____
Check: _____

8. $\frac{x}{3} - 2 = 4$
Step ①: $\frac{x}{3} =$ _____
Step ②: $x =$ _____
Check: _____

9. $2x + 9 = 3$

Step ①: $2x =$ _____

Step ②: $x =$ _____

Check: _____

10. $4x - 3 = 9$

Step ①: $4x =$ _____

Step ②: $x =$ _____

Check: _____

Answer Key (Teacher Copy)

Part A — One-Step Equations

1. $x + 5 = 9$ $x = 4$

2. $x - 3 = 7$ $x = 10$

3. $2x = 14$ $x = 7$

4. $\frac{x}{4} = 3$ $x = 12$

5. $x + 8 = 2$ $x = -6$

6. $x - 6 = -1$ $x = 5$

7. $3x = -12$ $x = -4$

8. $\frac{x}{5} = -2$ $x = -10$

9. $-x = 9$ $x = -9$

10. $x + 4 = -3$ $x = -7$

Part B — Two-Step Equations

1. $2x + 3 = 11$ $x = 4$

2. $3x - 5 = 10$ $x = 5$

3. $4x + 1 = 17$ $x = 4$

4. $2x - 7 = 1$ $x = 4$

5. $5x + 2 = -8$ $x = -2$

6. $3x - 4 = -13$ $x = -3$

7. $\frac{x}{2} + 3 = 7$ $x = 8$

8. $\frac{x}{3} - 2 = 4$ $x = 18$

9. $2x + 9 = 3$ $x = -3$

10. $4x - 3 = 9$ $x = 3$